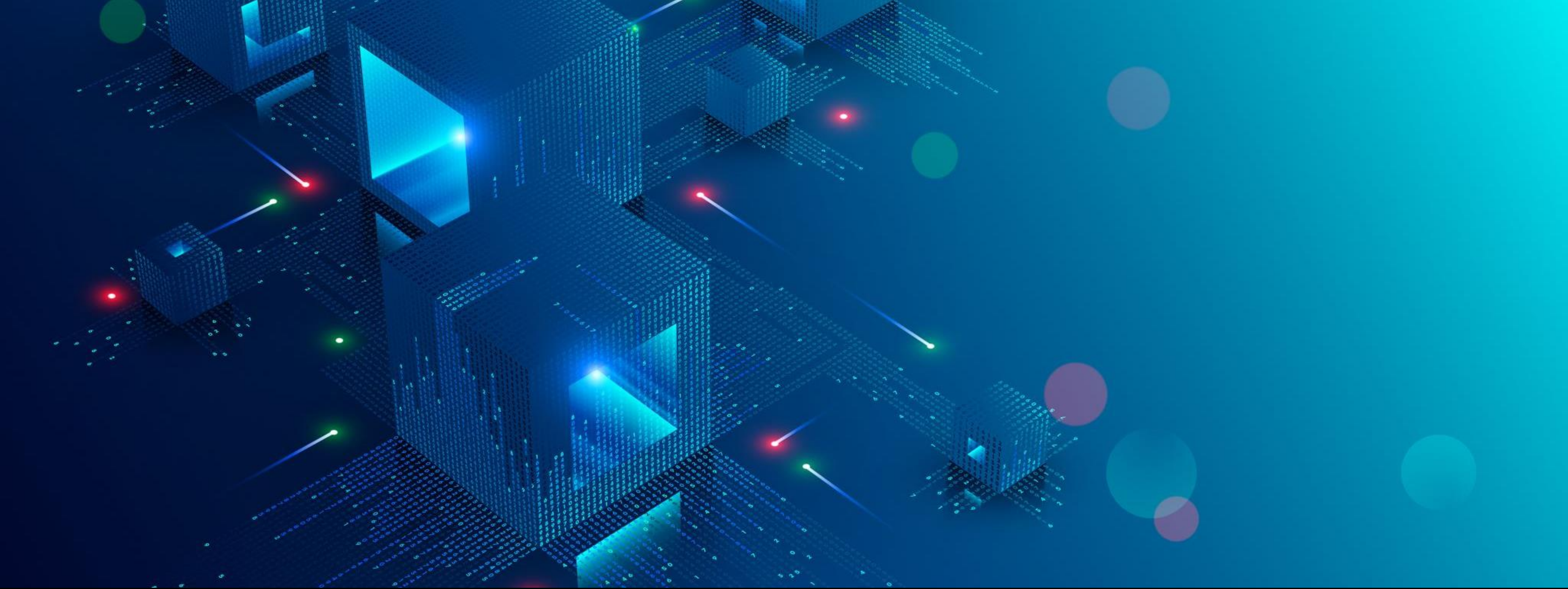
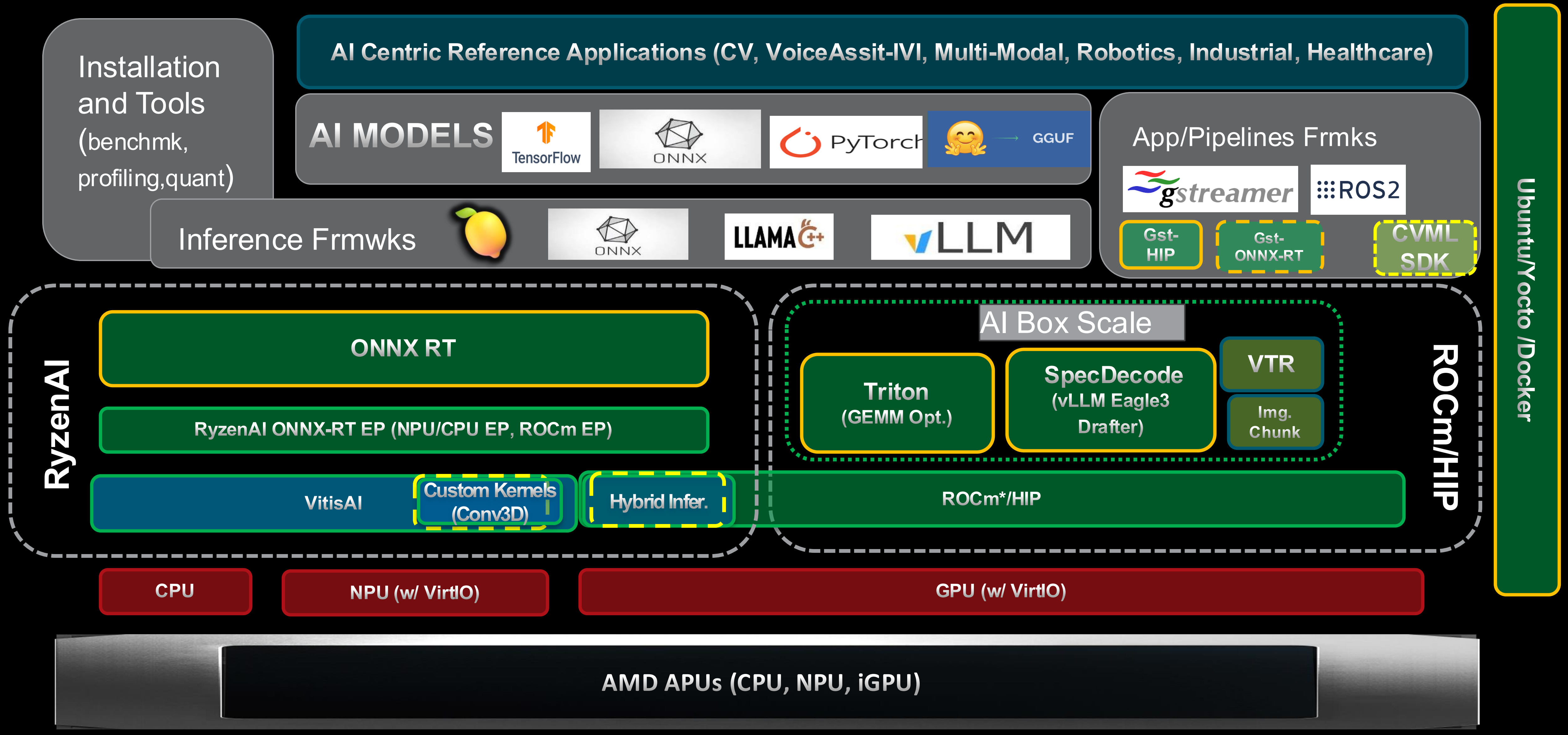


Physical AI SDK Architecture & Strategy

Zhaohui Yan
AECG/PAVS AI SW

Ryzen Embedded AI SW Stack (Physical AI SDK)



* AMD can port existing CUDA implementations into ROCm via HIP

Physical AI SDK – AI Models for Robotics Segment

Model Usage Category	Architecture	Model	KPIs (min goal)	Segment Priority
Instance Segmentation	ViT	MobileSAM	240fps	P0
Object detection, Pose Estimation, Segmentation	CNN	YoloV12	240fps	P0
Vision Transformer	ViT	CrossFormer	60fps	P2
3D Object Detection + Tracking	3D CNN	CenterPoint	30fps	P0
Vision Navigation Model	ViT	ViNT	30fps	P1
Vision Language Action	VLA	**OpenVLA**	TTFT:0.5s, 10Hz (Action)	P0
Vision Language Action	VLA	**Pi-0**	TTFT:0.5s, 10Hz (Action)	P0
Automatic Speech Recognition	Transformer	Insanely Fast Whisper	TTFT:0.1s, TPS:100tk/s	P0
Text to Speech	TTS	Xtts	TTFT:0.1s, TPS:100tk/s	P0
Multimodal LLM	VLM	LLaMA3.2-Vision 11B	TTFT:1s, TPS:25tk/s	P1
Vision Language Action	VLA	CogACT	TTFT:0.5s, 10Hz(Action)	P2
Object Pose model	CNN	GDR-Net	480fps	P1
LLM	LLM (DRL)	DeepSeek-R1 (Distilled, 7B)	TTFT:0.5s, TPS:50tk/s	P1
VLM	VLM	Qwen3-VL , Owen3-Omni-30B-A3B-Instruct	TTFT: 0.3s, TPS:60tk/s	P0

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Physical AI SDK – AI Models for Industrial Segment

Model Usage Category	Architecture	Model	KPI (min goal)	Segment Priority
Instance Segmentation	ViT	MobileSAM	240fps	P0
Object detection, Pose Estimation, segment.	CNN	YoloV12	240fps	P0
Image Classification	CNN	MobileNetV3	240fps	P2
Multi Object Tracking	CNN	DeepSort	240fps	P1
Image Classification	CNN	EfficientNetV2	240fps	P1
Multimodal LLM	VLM	LLaMA3.2-Vision 11B	TTFT:1s, TPS:25tk/s	P1
Optical Character Recognition (OCR)	CNN	EasyOCR	120fps	P1
Vision Language Model	VLM	CLIP	10fps (100msec)	P2
Multimodal LLM	VLM	InternVideo2.5	30fps , 20tk/s	P2
Anomaly Detection Localization	CNN	PaDiM	60fps	P1
Stereo Matching	CNN/RNN	RAFT Stereo	30fps	P1

OOB: Out of the Box.

Physical AI SDK – AI Models for Healthcare Segment

Model Usage Category	Architecture	Model	KPI (min goal)	Segment Priority
Instance Segmentation	ViT	MobileSAM	240fps	P0
Image Segmentation	3D CNN	3D U-NET, 3D V-NET, 3D AlexNet, 3DPSP Net	240fps	P1
Automatic Speech Recognition	Transformer	Insanely Fast Whisper	TTFT:0.1s, TPS:100tk/s	P0
Text to Speech	TTS	Xtts	TTFT:0.1s, TPS:100tk/s	P0
Object detection, Pose Estimation, Segmntt	CNN	YoloV12	240fps	P0
Image Classification	CNN	MobileNetV3	240fps	P2
Image Classification	CNN	EfficientNetV2	240fps	P1
Classification and Retrieval	VLM	MedSigLIP	10fps (100msec)	P1
LLM	LLM	DeepSeek-R1 (distill, 7B))	TTFT:0.5s, TPS:50tk/s	P1
Medical LLM	LLM	Med-PaLM & 2 (30B)	TTFT: 1s; TPS:10tk/s	P1
VQA	VLM	Qwen3-VL-4B	TTFT:0.1s, TPS:60tk/s	P0

Physical AI SDK – Inference Frameworks and Runtime

- AI Model Inference Frameworks Compatibility/Support by Physical AI SDK
  - vLLM : Compatible version 0.11.x with ROCm 7.1.x, Triton API, OpenAI API, Eagle3
  - ONNX-RT : Compatible version with RyzenAI 1.7/1.8 Releases
  - Llama.cpp : Compatible version with ROCm 7.1.x
  - Lemonade : Compatible version with RyzenAI 1.7/1.8 Releases, ROCm 7.1.x
- AI Workload Inference Runtime and AMD SW Stacks Supported by Physical AI SDK
  - RyzenAI (w/ VitisAI for NPU/CPU) : Releases 1.7/1.8, with Hybrid mode, custom kernels, for Linux (kernel 6.12vs6.14vs6.18)
  - ROCm : (formal support of STX-Halo): Release 7.1.x, with Linux Kernel 6.18
    - NOTE: Linux LTS Kernel convergence/migration: 6.12→6.18

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VLM Performance Optimization Components (via AI Box Scale)

Optimization Target	Mechanism	Descriptions	Uplift	Remark
TTFT/TPS	Triton Kernel Optimization (via vLLM/ROCm)	• Per-shape GEMM kernel optimization • Split-K GEMM (tiling) • Small GEMM to leaner-based large GEMM in MIOpen (Conv3D) • Windowed Attention via Triton • M-RoPE enablement for ViT in Qwen-VL support (1D→3D embedding) • ROCm INT8 kernels on iGPU	~50% overall	Applicable to non-Qwen-VL LLMs
TPS	AMD Speculative Decoding and Eagle Drafter in vLLM	• AMD (on AMD GPUs) trained Qwen2.5-VL SpecD Drafter (En and Ch) • Ported to vLLM Eagle 3 for size reduction • INT8 quantized drafter model (w8a8) • AMD finetuned drafter with specific automotive dataset (commercial)	~2x speedup (En)	Drafter MLOps
TTFT	Visual Token Reduction (VTR)	• Uses a "Token Reduce" ViT model from Qwen to reduce visual token count, importance sampling for image patch tokens • vLLM integration and vLLM compatible	~700 ms (input specific)	(Qwen license)
TTFT	Superframe, video preproc and Image Chunking	• Stich multiple images into a high-res mosaiced “super-frame” as input to Qwen-VL • Pre-processing images into video clips as input to Qwen-VL (achieved KPI for Geely) • Sub-divide high-res image (e.g. 960x540) into 2x2 or 4x4 sub-images as sequence of input to Qwen-VL (prompt engineering)	~20-30%	Innovations

Physical AI SDK – Tools & Infrastructure

- Performance Benchmarking and Profiling
  - Automated Model Benchmarking Framework
  - Selected Model Profling Anaylsis with OOB UX
- Workload Analysis and Quantization
  - AI Workload Model Per Layer “Hot Spot” Analysis via AI Analyzer
  - AWQ Quantization with OOB UX
- Installation and Deployment
  - Single installation for both ROCm and RyzenAI SW and toolchain
  - Docker container support for combined runtime env.
  - Pre-build, fully validated Docker images for easy custom eval
- Model Deployment
  - Hosting supported AI Model (including pre-quantized version, drafters) and datasets artifacts on AMD PAVS Harbor
- Model Training for Speculative Decoding
  - Hosting of Physical AI SDK supported VLMs/VLAs targeted drafter model training/re-training (PAVS+AIG)
  - Trained draft model evaluation support (accuracy and speedup)

Physical AI SDK – Pipeline Frameworks and Sample Apps (OOB)

- Gstreamer Framework for AI Media Analytics Pipelines (w/ ISV)
  - GstHIP
  - Gst-ONNX-RT
  - (opt.) VVAS (Leverage from AECG prior project)
- ROS2 and Integration for AI Workloads
  - Covered by PAVS Robotics SDK
- (opt.) CVML SDK (AMD G&E)
  - Working with ATG
- Sample Applications with OOB UX (w/ ISV-MCW, PSAS)
  - Concurrent Obj. detection with Realtime feed (2-4 concurrent video streams each 30fps@HD)
  - Concurrent ChatBot UX
  - Performance Benchmark with OOB for key AI Models
  - Robotics Simulation

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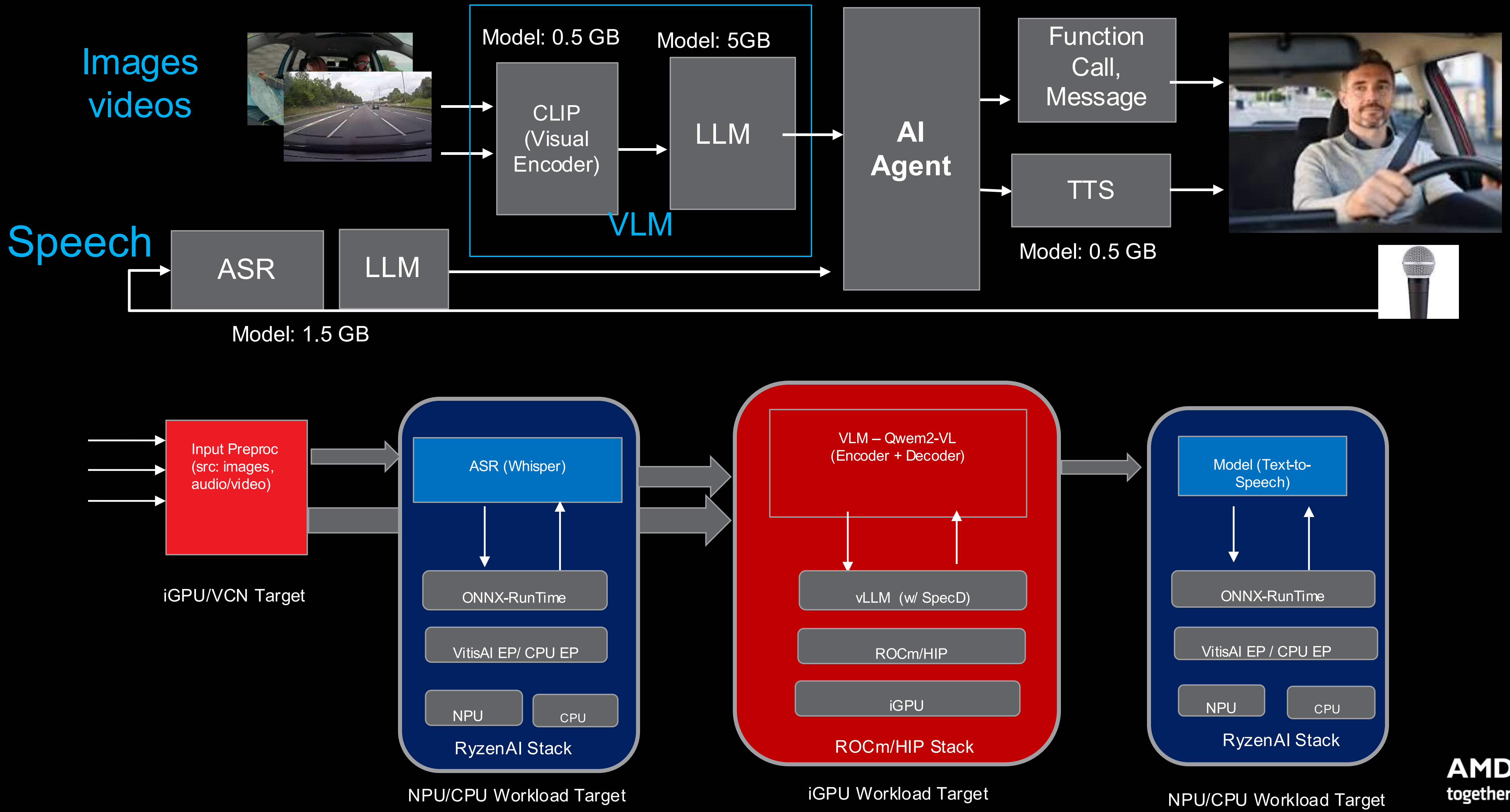
AI Inference Runtime Mapping – Heterogenous (In car voice asst. usage)



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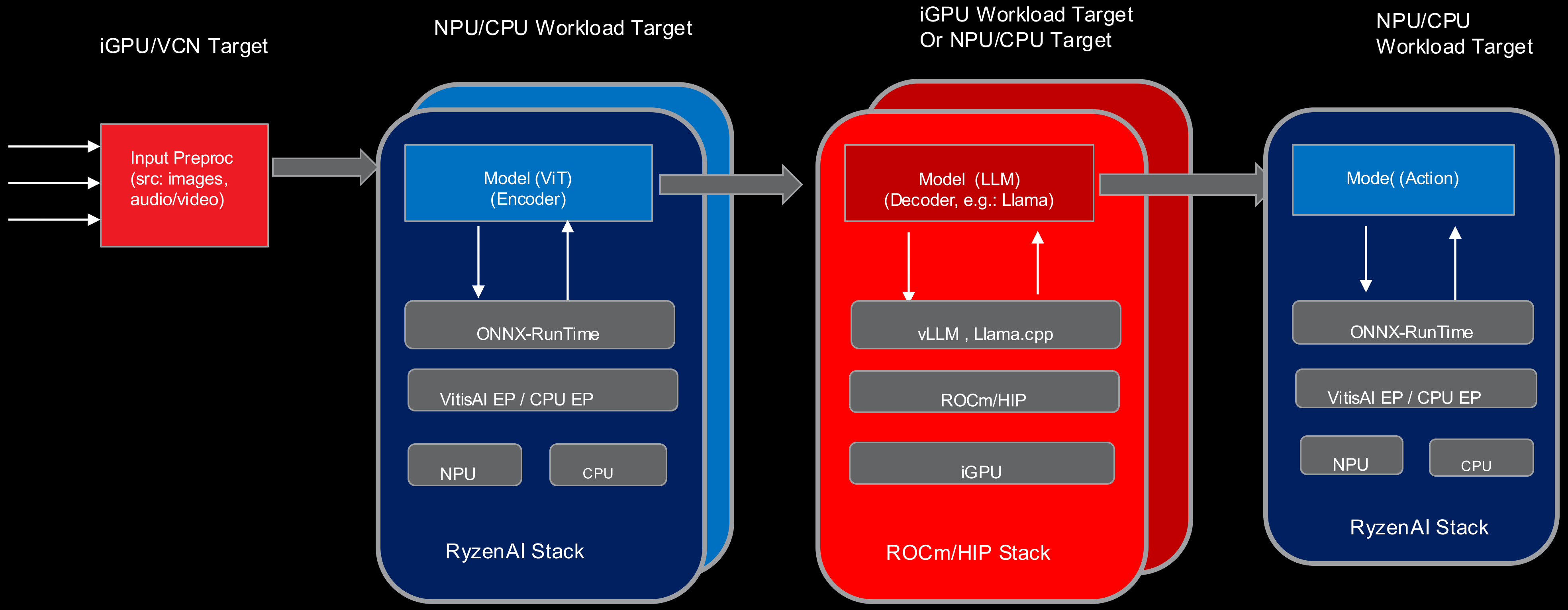
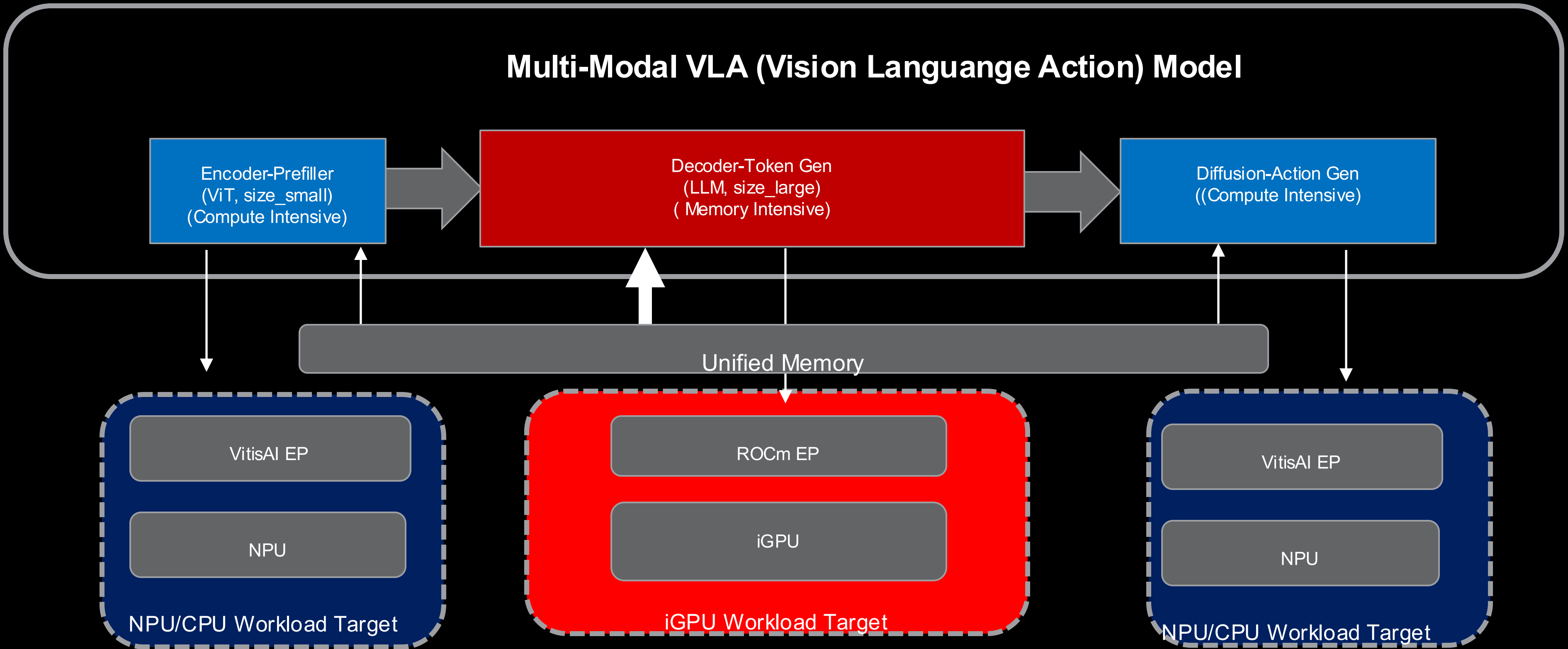
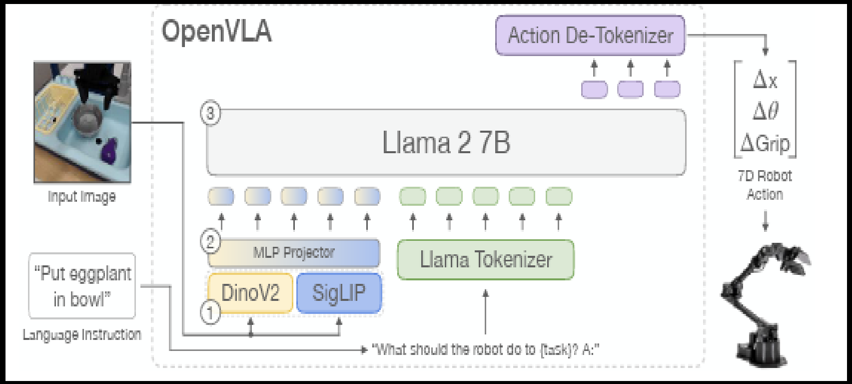
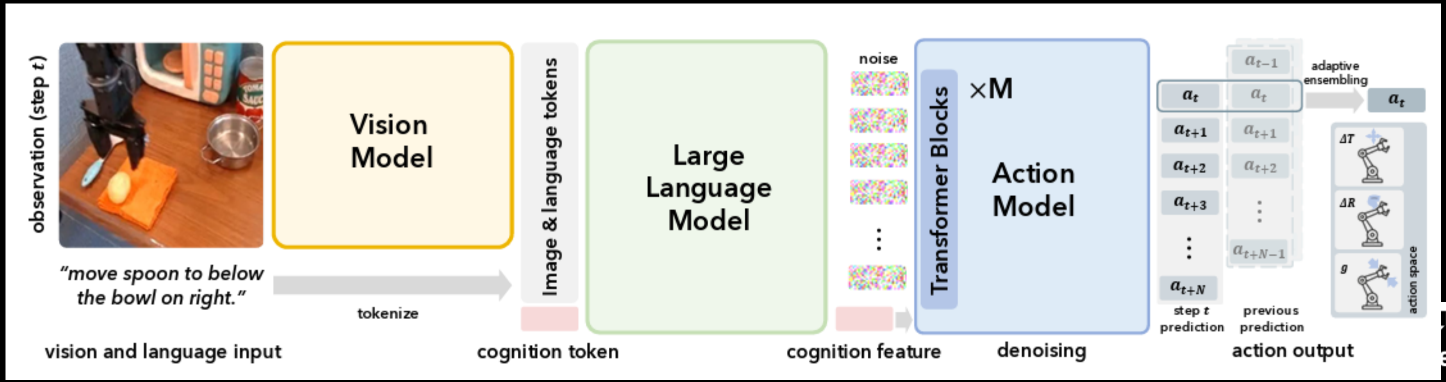
Agentic/Robotics (Ex. CogACT) Usage Mapping to Inference Pipeline



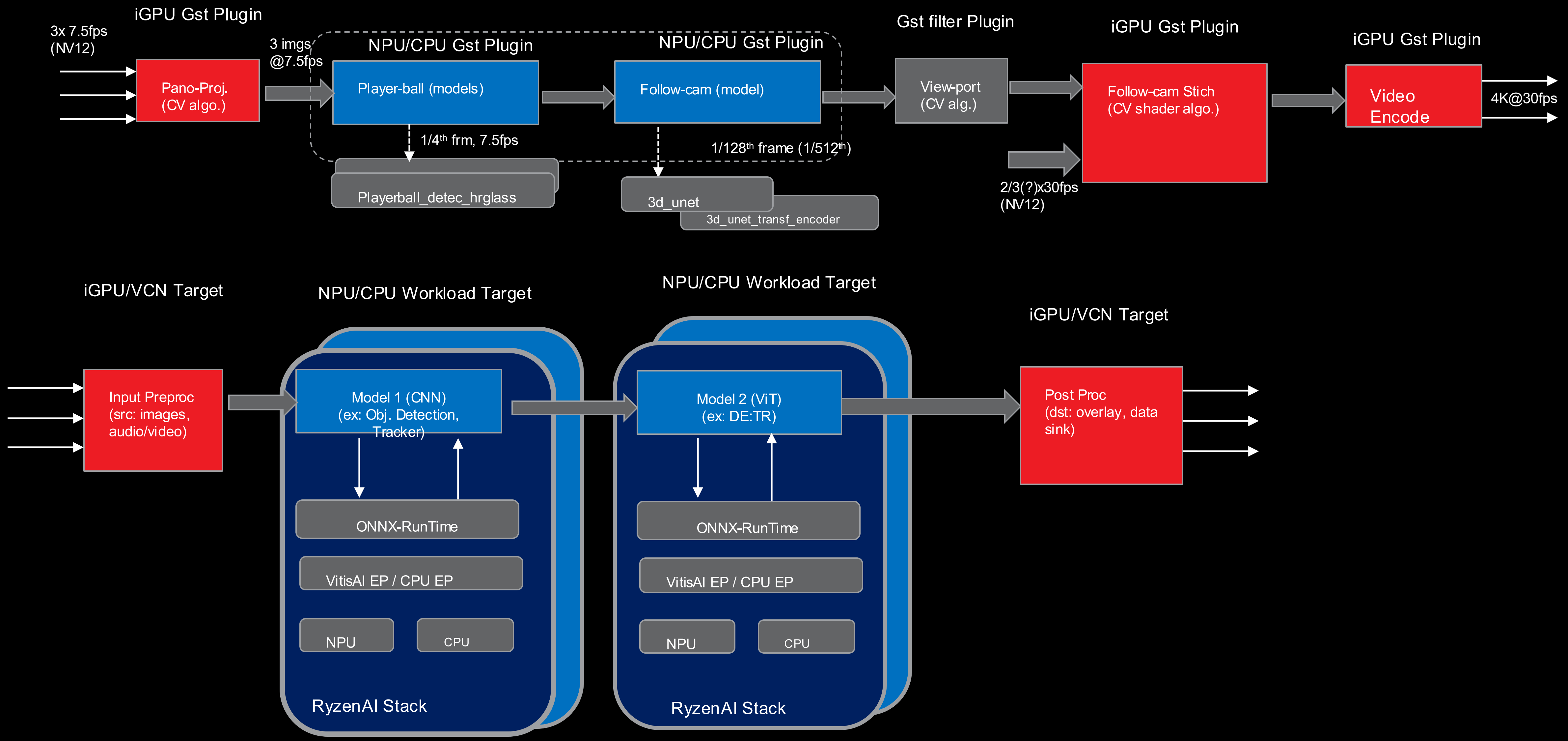
APU Hybrid Execution for Encoder-Decoder-Diffusion Models





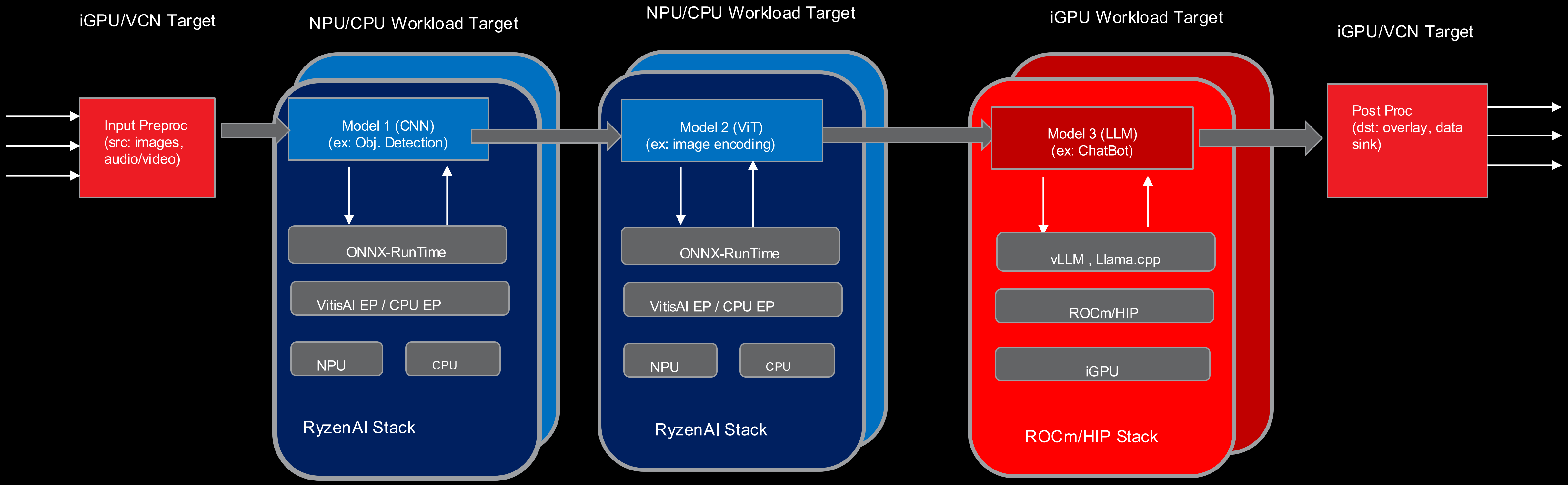


Industrial Application Usage Pipeline (Ex. Veo Follow-Cam)



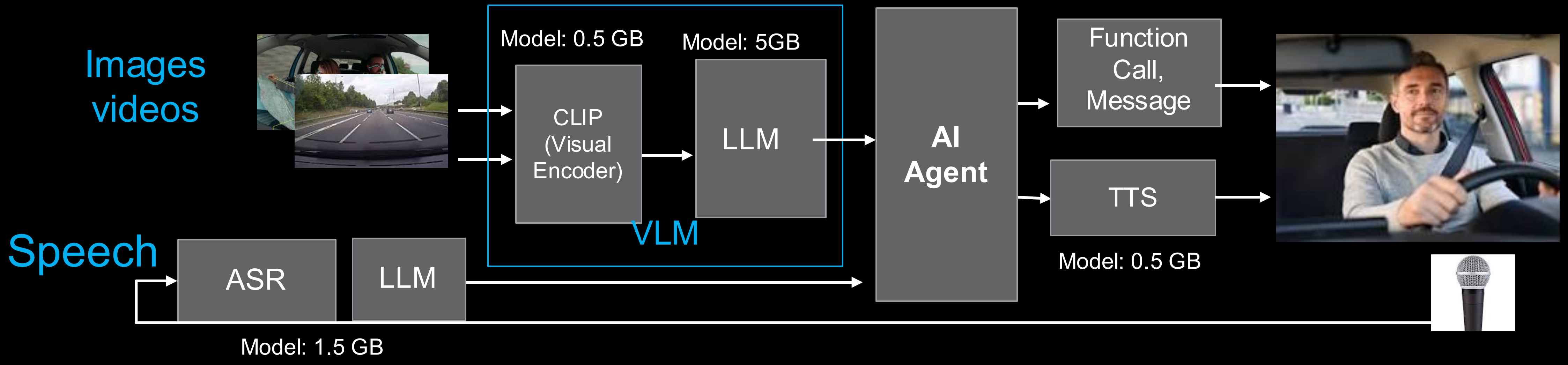
BACKUP

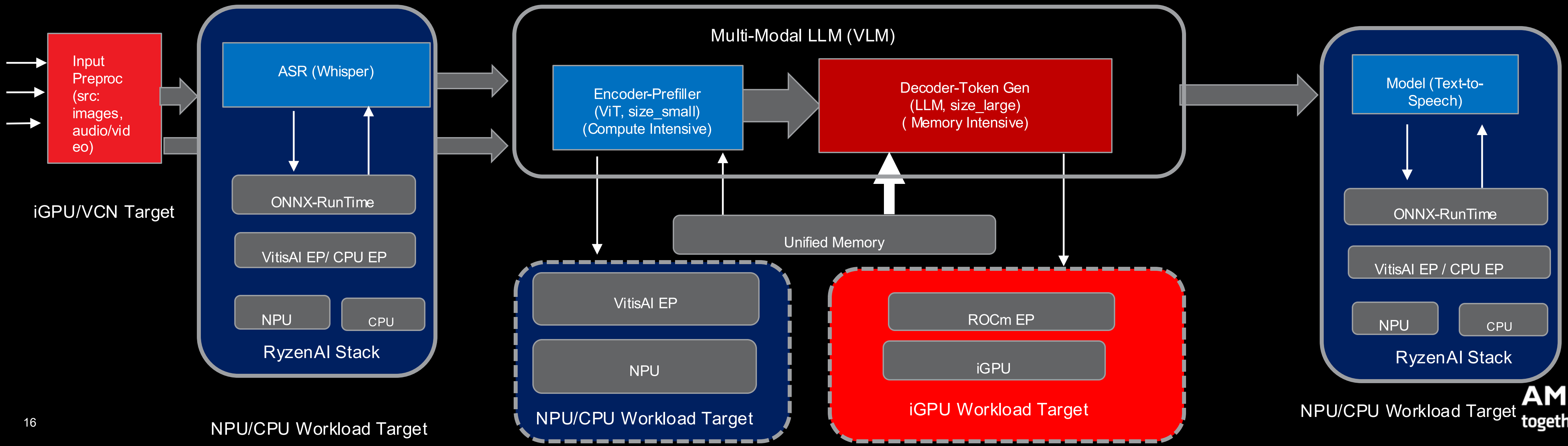
Embedded AI SW Stack Inference Runtime Flow - Generic



- iGPU – ROCm/HIP SW Stack
- NPU/CPU – RyzenAI SW Stack

In Vehicle Voice Assist. Usage to Inference Mapping – Hybrid (broader mkt, Krk2e)





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Vertical SW Stack – Building Blocks

Robotics <Product Name> Development Kit

Script
Documentation
Glue
Benchmark
SOC/CRB

Robotics SW Suite

- Robotic SDK
- Audio SDK
- Physical AI SDK
- Tools SDK

Foundation BSP

IVI <Product Name> Development Kit

Script
Documentation
Glue
Benchmark
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IVI SW Suite

- IVI SDK
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Foundation BSP

Industrial <Product Name> Development Kit

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Foundation BSP